

1. What is Raman Spectroscopy?

- a. Principle of Raman Spectroscopy: It is based on the interaction of light with a material's chemical bonds.
- b. Raman Spectrometer: Light is scattered off the sample and it is passed through a filter to remove particles from Rayleigh scattering, so the remnant light contains Raman scattered particles. It is then passed through a diffraction grating before entering the detector. Raman spectrum is then generated. Each peak and intensity in the spectrum provides some information about the sample.
- c. Light scattering during Raman Spectroscopy: Most of the incident light is scattered at the same wavelength as the incident wavelength and provides no meaningful information. This is called Rayleigh Scatter. However, a small amount of light scatters with a different wavelength than that of the incident light. This happens due to chemical structure of the sample and it is called Raman Scatter.
- d. Information provided by Raman Spectroscopy: Chemical structure and identity; Phase and polymorphism; Intrinsic stress and strain; Contamination and impurity.
- e. Types of samples analyzed by Raman Spectroscopy: Powders, gels, slurries; inorganic, organic and biological substances; pure chemicals, mixtures; metallic oxides; and others.

2. What is Fluorescence spectroscopy?

- a. Fluorescence: Electrons in a sample become excited when they absorb light and move from ground state to an excited state, that consists of various vibrational states. These excited electrons can transit to their ground state by emitting a photon and this process is called fluorescence.
- b. Fluorescence Excitation Spectrum: It is when the excitation wavelength is fixed and the emission wavelength is scanned to get a plot of intensity vs. emission wavelength.
- c. Information provided by Fluorescence spectroscopy: It provides the information about wavelengths at which a sample will absorb so as to emit at a single emission wavelength chosen for observation.
- d. Types of molecules exhibiting fluorescence: Chlorophyll, amino acids, organic dyes, lanthanides, phosphors, etc.

3. Which would you prefer and why?

I would prefer Raman spectroscopy for our rover. It has direct identification of specific minerals and organic compounds, which is highly beneficial for our biosignature identification point of view. Plus it does not need any sample preparation. Also in its setup, it does not always need a dark environment. On the other hand, Fluorescence spectroscopy has less specificity, due to overlap of spectrum regions for different samples. Moreover, it needs sample preparation sometimes. Since it depends on fluorescence, a dark environment is highly needed. Plus, it can only identify materials possessing the property of fluorescence, hence it can help us identify only a select few compounds.